Social Computing

Unit IV: Influence and Homophily

Syllabus:

Influence and Homophily: Measuring Assortativity, Influence, Homophily, Distinguishing Influence and Homophily: Shuffle test, Edge-Reversal Test, Randomization Test

- **Influence and Homophily**

Oct 2023

Q3) a) Explain the Homophily & Influence.

Ans. Homophily and Influence in Social Networks

Homophily and **Influence** are two important concepts used to explain why people in a social network often behave similarly.

1. Homophily

Definition

Homophily is the tendency of individuals to connect with and associate with people who are similar to themselves in terms of interests, age, education, profession, beliefs, or behavior.

Meaning:

"Birds of a feather flock together."

Characteristics

- People form friendships with others who share similar interests.
- Similarity exists **before** the relationship is formed.
- Common in social media platforms, communities, and organizations.

Example

- Students who enjoy gaming are more likely to become friends with other gamers.
- People interested in fitness often join fitness groups and connect with like-minded individuals.

Diagram

```
Sports Lovers          Sports Lovers
     A  -----------  B
```

```
Music Lovers          Music Lovers
     C  ----------  D
```

People with similar interests tend to connect with each other.

Advantages

- Easier communication.
- Stronger trust and understanding.
- Faster information sharing among similar people.

2. Influence

Definition

Influence is the process by which a person's opinions, behaviors, or decisions change because of interactions with others in the network.

Meaning:

People become similar **after** interacting with each other.

Characteristics

- Behavior spreads through social connections.
- One person's actions can affect others.
- Often observed in marketing, social media trends, and opinion formation.

Example

- A student starts using a new mobile app because all friends use it.
- People buy a product after seeing influencers recommend it.

Diagram

```
Person A (Uses App)
        |
        v
Person B --------> Starts Using App
```

Person A influences Person B's behavior.

Types of Influence

1. **Direct Influence** – One person directly affects another.
2. **Social Influence** – Influence through groups or communities.
3. **Viral Influence** – Information spreads rapidly across a network.

Difference Between Homophily and Influence

Homophily	Influence
Similar people become connected.	Connected people become similar.
Similarity exists before the relationship.	Similarity develops after interaction.
Based on shared characteristics.	Based on behavioral change.
Example: Gamers becoming friends.	Example: Friends starting the same game.

Relationship Between Homophily and Influence

In social networks, both phenomena often occur together:

1. People connect because they are similar (**Homophily**).
2. After connecting, they affect each other's behavior (**Influence**).

Example

Two students become friends because they both like technology (**Homophily**). Later, one student starts learning Python and encourages the other to learn it too (**Influence**).

Conclusion

Homophily explains **why people form connections**, while **Influence** explains **how connected people affect each other's behavior**. Both are fundamental concepts in social network analysis, recommendation systems, viral marketing, and data mining.

Q4) a) How to measure Influence of Homophily.

Ans. Measuring Influence and Homophily

In social network analysis, it is often difficult to determine whether people behave similarly because they **influence each other** or because they were already **similar before becoming connected (homophily)**. Researchers use several methods to measure and distinguish these effects.

1. Measuring Homophily

Homophily is measured by examining how similar connected individuals are compared to randomly selected individuals.

Homophily Index

$$H = \frac{\text{Observed Similar Connections}}{\text{Expected Similar Connections}}$$

- **H > 1** : Strong homophily exists.
- **H = 1** : No homophily.
- **H < 1** : Heterophily (people prefer dissimilar connections).

Example

Suppose in a social network:

- 80 friendships are between students of the same department.
- Random expectation is 50 friendships.

$$H = \frac{80}{50} = 1.6$$

Since (H > 1), strong homophily exists.

2. Measuring Influence

Influence is measured by observing behavioral changes over time after social interactions occur.

Influence Probability

$$P(I) = \frac{\text{Number of users adopting a behavior}}{\text{Number of exposed users}}$$

Example

- 100 users see a product recommendation from friends.
- 40 users purchase the product.

$$P(I) = \frac{40}{100} = 0.4$$

Thus, the influence probability is **40%**.

3. Distinguishing Influence from Homophily

Researchers compare user behavior at different times.

Time-Based Analysis

1. Measure similarity before friendship.
2. Measure similarity after friendship.
3. If similarity increases significantly after connection, influence is likely present.

Example

Time	Similarity Score
Before Friendship	0.45
After Friendship	0.80

Increase in similarity suggests social influence.

4. Common Techniques Used

A. Matching Method

- Compare connected users with similar but unconnected users.
- Helps isolate influence effects from homophily.

B. Regression Analysis

- Statistical models estimate how much behavior change is caused by friends.

C. Diffusion Models

Used to track how information spreads through a network.

Examples:

- **Independent Cascade Model**
- **Linear Threshold Model**

D. Randomized Experiments

- Some users receive information while others do not.
- Differences in behavior measure influence directly.

Diagram

```
   Homophily
(Similarity First)
Student A ------ Student B
(Both like AI)
       ↓
   Friendship
       ↓
   Influence
(Behavior Change)
```

```
Student A ---> Learns Python
                 ↓
Student B ---> Learns Python
```

Applications

- Viral marketing
- Product recommendation systems
- Social media analysis
- Information diffusion studies
- Community detection
- Data mining and machine learning

Conclusion

Homophily is measured by the similarity of connected users, while **Influence** is measured by changes in behavior after interactions. To measure the influence of homophily, researchers compare users' similarities before and after forming connections using methods such as homophily indices, influence probabilities, matching techniques, regression models, and diffusion analysis.

- **Measuring Assortativity**

Apr 2023

Q4) a) What is measuring assortativity.

Ans. Measuring Assortativity

Definition

Assortativity is a measure used in social network analysis to determine the tendency of nodes (people, organizations, etc.) to connect with other nodes that are similar to them.

It quantifies the degree of **homophily** in a network. If similar nodes tend to connect with each other, the network is said to be **assortative**.

Assortativity Coefficient

The assortativity coefficient (r) measures the correlation between connected nodes.

$$-1 \leq r \leq 1$$

Where:

- **(r = 1)** : Perfect assortative mixing (similar nodes connect together).

- **(r = 0)** : No preference in connections.
- **(r = -1)** : Perfect disassortative mixing (dissimilar nodes connect together).

Types of Assortativity

1. Attribute Assortativity

Measures similarity based on attributes such as:

- Age
- Gender
- Profession
- Education
- Interests

Example: Engineers mostly connect with engineers.

2. Degree Assortativity

Measures whether nodes with similar numbers of connections (degree) tend to connect.

Example:

- Highly connected users connect with other highly connected users.
- Low-degree users connect with low-degree users.

Formula for Degree Assortativity

$$r = \frac{\sum_i j_i k_i - M^{-1} \sum_i j_i \sum_i k_i}{\sqrt{\left[\sum_i j_i^2 - M^{-1}(\sum_i j_i)^2\right]\left[\sum_i k_i^2 - M^{-1}(\sum_i k_i)^2\right]}}$$

Where:

- j_i and k_i are the degrees of nodes at the ends of edge (i).
- (M) is the total number of edges.

This formula calculates the Pearson correlation coefficient between the degrees of connected nodes.

Example

Consider a social network:

```
A (Engineer) ---- B (Engineer)

C (Doctor) ------ D (Doctor)
```

```
A (Engineer) ---- C (Doctor)
```

There are more connections between people of the same profession than between different professions.

Therefore:

$r > 0$

The network shows **positive assortativity**.

Interpretation

Assortativity Coefficient (r)	Interpretation
$r = 1$	Perfect assortative network
$0 < r < 1$	Positive assortativity
$r = 0$	Random mixing
$-1 < r < 0$	Negative assortativity
$r = -1$	Perfect disassortative network

Applications of Assortativity

- Social network analysis
- Community detection
- Recommendation systems
- Data mining
- Epidemic spreading analysis
- Online social media studies

Conclusion

Assortativity is a metric that measures how strongly similar nodes connect with one another in a network. It is represented by the **assortativity coefficient ((r))**, whose value ranges from **-1 to +1**. Positive values indicate **homophily (similar nodes connect)**, while negative values indicate **heterophily (dissimilar nodes connect)**. It is widely used to analyze the structure and behavior of social networks.

Nov 2024

Q3) a) What is Assortativity? Explain any one technique to measure assortativity.

Ans. Assortativity

Definition

Assortativity is a measure in network analysis that indicates the tendency of nodes (people, organizations, web pages, etc.) to connect with other nodes that are similar to them. It helps determine whether similar nodes are more likely to be connected than dissimilar nodes.

In social networks, assortativity is often used to measure **homophily**, i.e., the tendency of people with similar characteristics to form relationships.

Types of Assortativity

1. **Attribute Assortativity** – Based on attributes such as age, gender, occupation, education, etc.
2. **Degree Assortativity** – Based on the number of connections (degree) of nodes.

Technique to Measure Assortativity: Degree Assortativity Coefficient

One common technique is the **Degree Assortativity Coefficient**, which measures whether nodes with similar degrees tend to connect.

Formula

$$r = \frac{\sum_i j_i k_i - M^{-1} \sum_i j_i \sum_i k_i}{\sqrt{\left[\sum_i j_i^2 - M^{-1}(\sum_i j_i)^2\right]\left[\sum_i k_i^2 - M^{-1}(\sum_i k_i)^2\right]}}$$

Where:

- j_i = degree of one node of edge (i)
- k_i = degree of the other node of edge (i)
- (M) = total number of edges
- (r) = assortativity coefficient

Interpretation

Value of r	Meaning
$r = 1$	Perfect assortative mixing
$0 < r < 1$	Positive assortativity
$r = 0$	No assortativity (random mixing)
$-1 < r < 0$	Negative assortativity
$r = -1$	Perfect disassortative mixing

Example

Consider the following network:

```
A (Degree 4) ------ B (Degree 5)
C (Degree 1) ------ D (Degree 2)
```

Here:

- High-degree nodes connect with other high-degree nodes.
- Low-degree nodes connect with other low-degree nodes.

This network exhibits **positive assortativity** ((r > 0)).

Diagram

```
High Degree Nodes
A(4) -------- B(5)

Low Degree Nodes
C(1) -------- D(2)
```

Since nodes tend to connect with others having similar degrees, the assortativity coefficient will be positive.

Applications

- Social network analysis
- Community detection
- Recommendation systems

- Information diffusion studies
- Data mining and graph analytics

Conclusion

Assortativity measures the preference of nodes to connect with similar nodes in a network. One widely used technique is the **Degree Assortativity Coefficient**, which calculates the correlation between the degrees of connected nodes. A positive coefficient indicates that similar nodes tend to connect, while a negative coefficient indicates connections between dissimilar nodes.

- **Homophily**

Apr 2024

Q4) b) What is Homophily? How to measure Homophily.

Ans. Homophily and Its Measurement

What is Homophily?

Homophily is the tendency of individuals to associate and form relationships with people who are similar to themselves in terms of age, gender, education, profession, interests, beliefs, or behavior.

In simple words:

"Birds of a feather flock together."

Homophily is a fundamental concept in social computing and social network analysis because it explains why people with similar characteristics tend to connect and interact.

Examples

- Students from the same department often become friends.
- People with similar hobbies join the same online communities.
- Professionals in the same field are more likely to connect on social networking sites.

Diagram

```
Engineering Students
A -------- B
Medical Students
C -------- D
```

Here, people with similar backgrounds form connections, showing homophily.

Types of Homophily

1. Status Homophily

Based on social characteristics such as:

- Age
- Gender
- Education
- Occupation
- Religion

Example: Doctors interacting mainly with other doctors.

2. Value Homophily

Based on shared beliefs, interests, attitudes, and values.

Example: People interested in photography joining the same online group.

How to Measure Homophily?

1. Homophily Index

The Homophily Index compares the observed number of similar connections with the expected number of similar connections.

Formula

$$H = \frac{\text{Observed Similar Connections}}{\text{Expected Similar Connections}}$$

Interpretation

Value of H	Meaning
H > 1	Strong homophily
H = 1	No homophily
H < 1	Heterophily (preference for dissimilar connections)

Example

Suppose:

- Observed friendships among students of the same department = 80
- Expected friendships = 50

$$H = \frac{80}{50} = 1.6$$

Since H > 1, strong homophily exists.

2. Assortativity Coefficient

Assortativity measures how strongly similar nodes connect in a network.

$$-1 \leq r \leq 1$$

Where:

- (r = 1): Perfect homophily
- (r = 0): Random connections
- (r = -1): Dissimilar nodes prefer connecting

Example

If engineers mostly connect with engineers and doctors with doctors, the network has positive assortativity.

3. Similarity-Based Measures

Homophily can also be measured using similarity metrics such as:

- Cosine Similarity
- Jaccard Similarity
- Pearson Correlation

These compare users based on shared attributes or interests.

Jaccard Similarity Formula

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

A higher value indicates stronger similarity and greater homophily.

Applications of Homophily

- Social network analysis
- Friend recommendation systems
- Community detection
- Targeted advertising
- Data mining
- Information diffusion studies

Conclusion

Homophily is the tendency of people to connect with others who are similar to themselves. It plays an important role in the formation of social networks. Homophily can be measured using techniques such as the **Homophily Index**, **Assortativity Coefficient**, and **Similarity Measures**. Higher values of these measures indicate stronger similarity and stronger homophily among network members.

- **Influence**

Apr 2024

Q3) a) What is Influence and how to Measure Influence?

Nov 2024

Q3) b) What is influence? Explain how to measure influence in social computing.

Ans. Influence in Social Computing and Its Measurement

What is Influence?

Influence in social computing refers to the ability of an individual, group, or entity in a social network to affect the opinions, behaviors, decisions, or actions of others.

In social networks, influence occurs when one user's actions, recommendations, or opinions lead other users to adopt similar behaviors.

Examples

- A social media influencer promotes a product, and followers purchase it.
- A friend recommends a movie, and others decide to watch it.
- A user's post goes viral, causing many people to share the same content.

Diagram

```
User A (Influencer)
        |
        v
User B ----> Adopts Behavior
        |
        v
User C ----> Adopts Behavior
```

Here, User A influences Users B and C.

Measuring Influence in Social Computing

Several techniques are used to measure influence in social networks.

1. Degree Centrality

Measures influence based on the number of direct connections a user has.

Formula

$$DC(v) = \frac{\deg(v)}{N - 1}$$

Where:

- deg(v) = number of connections of node v
- N= total number of nodes

Interpretation

- Higher degree → greater potential influence.
- Users with many followers/friends are often influential.

2. Influence Probability

Measures how often users adopt a behavior after being exposed to it.

Formula

$$P(I) = \frac{\text{Number of users adopting behavior}}{\text{Number of exposed users}}$$

Example

- 100 users see a product recommendation.
- 40 users buy the product.

$$P(I) = \frac{40}{100} = 0.4$$

Influence probability = **40%**.

3. PageRank-Based Influence

Uses network structure to identify influential users.

- A user is influential if they are connected to other influential users.
- Widely used by social media and search engines.

Example

A celebrity followed by many important users will receive a higher influence score.

4. Information Diffusion Analysis

Measures how information spreads through the network.

Metrics

- Number of shares
- Number of retweets
- Cascade size
- Reach of information

Greater spread indicates greater influence.

5. Independent Cascade Model

A popular influence measurement model.

Process

1. An active user influences neighbors.
2. Each neighbor adopts behavior with a certain probability.
3. Newly influenced users continue spreading information.

Used in:

- Viral marketing
- Social media analysis
- Product recommendation systems

6. Linear Threshold Model

A user adopts a behavior when influence from neighbors exceeds a threshold.

Example

If 3 out of 5 friends use an app and the threshold is 60%, the user may also adopt the app.

Applications of Influence Measurement

- Viral marketing
- Recommendation systems
- Social media analytics
- Opinion mining
- Information diffusion studies
- Community detection
- Advertising campaigns

Conclusion

Influence is the ability of a user to affect the behavior, opinions, or decisions of others in a social network. It can be measured using techniques such as **Degree Centrality, Influence Probability, PageRank, Information Diffusion Analysis, Independent Cascade Model, and Linear Threshold Model**. Measuring influence helps organizations identify key users and understand how information spreads in social computing environments.

- ♦ **Distinguishing Influence and Homophily**

Apr 2023

Q3) a) Distinguish influence and Homophily.

Ans. Difference Between Influence and Homophily

Aspect	Influence	Homophily
Definition	Influence is the process by which a person's behavior, opinions, or decisions are affected by others in a social network.	Homophily is the tendency of individuals to form connections with people who are similar to themselves.
Reason for Similarity	People become similar **after** interacting.	People are similar **before** forming a connection.
Focus	Behavioral change due to social interactions.	Formation of relationships based on similarity.
Cause	Social pressure, recommendations, communication, and interaction.	Shared interests, values, demographics, or characteristics.
Direction	Causes changes in attitudes or behavior over time.	Causes people to choose similar friends or connections.
Time Factor	Similarity develops after the relationship is established.	Similarity exists before the relationship is established.
Network Effect	Information, ideas, or behaviors spread through the network.	Similar individuals cluster together in the network.
Example	A student starts using Instagram because all friends use it.	Students who already like social media become friends.
Measurement	Measured using influence probability, diffusion models, PageRank, and centrality measures.	Measured using homophily index, assortativity coefficient, and similarity measures.
Outcome	Leads to adoption of behaviors, products, or opinions.	Leads to formation of communities and social groups.

Diagram

```
          HOMOPHILY
   (Similarity Before Connection)

 Student A (Likes Gaming)
```

```
           |
           |
Student B (Likes Gaming)

           ↓ Friendship


           INFLUENCE
  (Similarity After Connection)

Student A (Uses New App)
           |
           ↓
Student B Starts Using New App
```

Example

Consider two students, A and B:

Homophily

- A and B both enjoy programming.
- Because of this common interest, they become friends.

Influence

- After becoming friends, A learns Python.
- B is motivated by A and also starts learning Python.

Thus:

- **Homophily explains why they became friends.**
- **Influence explains how one friend's behavior affected the other.**

Conclusion

Homophily and **Influence** are two key concepts in social computing and social network analysis. **Homophily** explains the formation of connections based on similarity, whereas **Influence** explains how connected individuals affect each other's behavior, opinions, and decisions over time.

- ♦ **Shuffle test**

Oct 2023

Q3) b) What is a shuffle test designed for in the context of social influence? Describe how it is performed.

Ans. Shuffle Test in Social Influence

What is a Shuffle Test?

A **Shuffle Test** is a statistical technique used in social network analysis to determine whether observed behavioral similarities among connected individuals are due to **social influence** or simply due to **homophily** (similar people connecting with each other).

The main purpose of the shuffle test is to verify whether influence actually exists in a network by comparing real data with randomly shuffled data.

Purpose of a Shuffle Test

The shuffle test is designed to:

- Distinguish **social influence** from **homophily**.
- Check whether behavior spreads through social connections.
- Determine if observed correlations are statistically significant.
- Validate influence models used in social computing.

Example

Suppose friends in a social network start using a new mobile app.

The similarity in app usage may be because:

1. **Homophily:** Friends already had similar interests.
2. **Influence:** One friend persuaded others to use the app.

A shuffle test helps determine which explanation is more likely.

How is a Shuffle Test Performed?

Step 1: Collect Network Data

Gather:

- Users (nodes)
- Friendships/connections (edges)
- Behavioral data over time

Example:

User	Adopted App
A	Yes
B	Yes
C	No
D	Yes

Step 2: Measure Actual Influence

Calculate the observed influence effect.

Example:

- 50 friends adopted an app after their friends adopted it.
- Observed influence score = 50.

Step 3: Shuffle the Data

Randomly rearrange (shuffle) the timing or assignment of behaviors among users while keeping the network structure unchanged.

Example:

Original:

```
A → Adopts App
B → Adopts After A
```

After Shuffling:

```
A → No Adoption
B → Adoption Assigned Randomly
```

The friendship network remains the same, but behavior labels or timestamps are randomized.

Step 4: Recalculate Influence

Compute the influence score on the shuffled dataset.

Repeat the shuffling many times (e.g., 1000 iterations) to generate a distribution of influence scores.

Step 5: Compare Results

Compare:

Observed Influence Score

with

Average Shuffled Influence Score

Interpretation

- **Observed score much higher than shuffled scores:** Strong evidence of social influence.
- **Observed score similar to shuffled scores:** Similarity is likely due to homophily or chance.

Diagram

```
Real Network
A ---- B ---- C

Behavior Spread
A adopts → B adopts → C adopts

        ↓

Shuffle Behaviors Randomly

A ---- B ---- C

B adopts, A no adoption

        ↓

Compare Results
```

Advantages of Shuffle Test

- Simple and easy to implement.
- Separates influence from homophily.
- Provides statistical validation.
- Widely used in social network research.

Applications

- Social media analysis
- Viral marketing
- Information diffusion studies
- Product adoption analysis
- Social computing research

Conclusion

A **Shuffle Test** is a statistical method used to detect **social influence** by comparing observed behavioral patterns with patterns obtained after randomly shuffling behavior data. If the real influence effect is significantly greater than the shuffled effect, it indicates that behavior is spreading through **social influence** rather than being explained solely by **homophily** or random chance.

- ♦ **Edge-Reversal Test**

Apr 2023

Q3) b) Explain in detail edge-reversal test.

Oct 2023

Q4) b) Describe how the edge reversal test works. What is it used for?

Ans. Edge-Reversal Test

What is an Edge-Reversal Test?

The **Edge-Reversal Test** is a method used in social network analysis to determine whether observed behavioral similarities between connected users are caused by **social influence** rather than **homophily** or other factors.

It examines the **direction of relationships (edges)** in a network and checks whether influence depends on the direction of the connection.

Purpose of the Edge-Reversal Test

The test is used to:

- Detect the presence of **social influence**.
- Distinguish influence from **homophily**.
- Verify whether influence follows the direction of social ties.
- Validate causal relationships in social networks.

Example

Consider a social media network:

```
A   ---->  B
(Follows)
```

If A influences B, then B's behavior should be affected by A's actions.

The edge-reversal test checks what happens when the direction is reversed:

```
B   ---->  A
```

If influence disappears or becomes much weaker after reversing the edge, it suggests that the original influence was genuine.

How Does the Edge-Reversal Test Work?

Step 1: Observe the Original Network

Consider a directed social network:

```
A ----> B
```

- A is connected to B.
- Measure the influence of A on B.

Suppose:

- A adopts a new app.
- B later adopts the same app.

This suggests possible influence.

Step 2: Calculate Influence in Original Direction

Measure influence using adoption rates, diffusion probability, or another influence metric.

Example:

```
A ----> B

Influence Score = 0.70
```

Step 3: Reverse the Edge

Reverse the direction of the relationship:

```
B ----> A
```

The network structure remains the same except for the edge direction.

Step 4: Recalculate Influence

Measure influence again using the reversed network.

Example:

```
B ----> A
Influence Score = 0.10
```

Step 5: Compare Results

Compare:

Original Influence Score

with

Reversed Influence Score

Interpretation

Result	Interpretation
Original score ≫ Reversed score	Strong evidence of social influence
Original score ≈ Reversed score	Similarity may be due to homophily or common external factors
No significant difference	Weak evidence for influence

Example

Consider Twitter-like followers:

```
Celebrity ----> Followers
```

Suppose:

- Many followers start using a hashtag after the celebrity uses it.
- Influence score = 0.80

After reversing:

```
Followers ----> Celebrity
```

Influence score becomes 0.05.

Since influence disappears after reversal, the original relationship likely reflects genuine social influence.

Diagram

```
Original Network
A ----> B
Influence = 0.70
       ↓ Reverse Edge
B ----> A
Influence = 0.10
Result:
0.70 > 0.10
⇒ Evidence of Influence
```

Advantages

- Simple method for validating influence.
- Helps separate influence from homophily.
- Useful for directed social networks.
- Provides evidence of causal relationships.

Applications

- Social media analysis
- Viral marketing
- Information diffusion studies
- Product recommendation systems
- Social computing research
- Network science

Conclusion

The **Edge-Reversal Test** is a technique used to verify social influence in a network by reversing the direction of connections and comparing influence scores. If influence significantly decreases after reversing the edges, it indicates that the original effect was due to **social influence** rather than **homophily** or random similarity. It is widely used in social computing and network analysis to study how behaviors, information, and opinions spread through social networks.

- ♦ **Randomization Test**

Apr 2023

Q4) b) Explain in detail Randomization test.

Apr 2024

Q3) b) Explain Randomization Test with example.

Ans. Randomization Test

Definition

A **Randomization Test** (also called a **Permutation Test**) is a statistical method used to determine whether an observed pattern or relationship in a social network is significant or could have occurred by chance.

In social computing, it is commonly used to:

- Measure **social influence**.
- Detect **homophily**.
- Validate network patterns.
- Test the significance of information diffusion.

The basic idea is to compare the **actual observed result** with results obtained after randomly rearranging (randomizing) the data many times.

Steps of a Randomization Test

Step 1: Calculate the Observed Statistic

Measure the quantity of interest from the real network.

Example:

- 60 out of 100 users adopt a product after their friends adopt it.
- Observed influence score = 60%.

Step 2: Randomize the Data

Randomly shuffle:

- User behaviors,
- Attributes, or
- Adoption times,

while keeping the network structure unchanged.

Step 3: Recalculate the Statistic

Calculate the influence score for the randomized network.

Example:

Trial	Influence Score
1	30%
2	35%
3	28%
4	33%
5	31%

Average randomized influence = 31.4%.

Step 4: Repeat Many Times

Perform hundreds or thousands of randomizations to generate a distribution of scores.

Step 5: Compare Results

Compare the observed score with the randomized scores.

- If the observed value is much larger than random values, the effect is statistically significant.
- Otherwise, the pattern may have occurred by chance.

Example

Product Adoption in a Social Network

Suppose a company wants to know whether friends influence each other to buy a new smartphone.

Real Data

A → B → C → D

- A buys the phone.
- Later B, C, and D also buy it.

Observed adoption rate:

$$\frac{4}{5} = 0.80$$

or **80%**.

Randomized Data

After randomly assigning purchases:

```
A (No)
B (Yes)
C (No)
D (Yes)
```

Adoption rates from several random trials:

Trial	Adoption Rate
1	35%
2	40%
3	32%
4	38%
5	36%

Average random adoption rate:

36.2%

Comparison

Measure	Value
Observed Adoption Rate	80%
Randomized Adoption Rate	36.2%

Since:

80% > 36.2%

the observed behavior is unlikely due to chance, suggesting **social influence**.

Diagram

```
Original Network
A → B → C → D
80% Adoption
        ↓ Randomization
A → B → C → D
36% Adoption
        ↓
80% >> 36%
⇒ Significant Influence
```

Applications

- Social influence detection
- Homophily analysis
- Viral marketing
- Recommendation systems
- Information diffusion studies
- Social network research

Advantages

- Does not require strong statistical assumptions.
- Easy to understand and implement.
- Works well with complex network data.
- Provides reliable significance testing.

Conclusion

A **Randomization Test** is a statistical technique that evaluates whether an observed social network pattern is significant by comparing it with results obtained from randomly shuffled data. If the observed result is much stronger than the randomized results, it indicates that the effect is likely due to **social influence or homophily** rather than random chance. It is widely used in social computing, network analysis, and data mining.